

NanoGT Match Report: Achromatopsia

Disease: Achromatopsia (ORPHA:49382)

Primary gene: CNGB3

Gene CDS: 2427 bp

Inheritance: Autosomal recessive

Target tissues: retina

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	CPCB-RPE1	AAV8	7.1/10	[YELLOW] Medium	phase2/3
2	Luxturna	AAV2	6.7/10	[YELLOW] Medium	approved
3	Skysona	LV	6.1/10	[YELLOW] Medium	approved
4	Libmeldy	LV	5.7/10	[YELLOW] Medium	approved
5	Strimvelis	LV	5.7/10	[YELLOW] Medium	approved

Match #1: CPCB-RPE1

Precedent disease: Achromatopsia

Vector: AAV8

Tissue target: retina/photoreceptor

Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.70	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.06	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 2427bp / cargo 4700bp (52% utilized)
- Precedent target match: retina
- Both membrane proteins
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: retinal phototransduction
- Approval status: phase2/3
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010

Match #2: Luxturna

Precedent disease: Leber congenital amaurosis type 2

Vector: AAV2

Tissue target: retina/RPE

Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	0.50	2.0	Same se-creted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	0.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
TOTAL (normalised)	6.67	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 2427bp / cargo 4700bp (52% utilized)
- Vector tropism plus precedent target match: retina
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: retinal_visual_cycle
- Approval status: approved
- Vector immunogenicity (AAV2): very high (~55%) — majority of patients may be ineligible; major trial design challenge
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010

Match #3: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
TOTAL (normalised)	6.11	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 2427bp / cargo 8000bp (30% utilized)
- No tissue overlap (disease: ['retina'], vector: ['hematopoietic'])
- Both membrane proteins
- LOF inheritance — compatible for gene replacement
- Different pathway (retinal_phototransduction vs peroxisomal)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity

- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- β 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010

Match #4: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 5.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
TOTAL (normalised)	5.72	10.0	Raw sum / 18×10

Rationale

- Gene CDS 2427bp / cargo 8000bp (30% utilized)
- No tissue overlap (disease: ['retina'], vector: ['hematopoietic'])
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (retinal_phototransduction vs leukodystrophy)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF- β 2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010

Match #5: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 5.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
TOTAL (normalised)	5.72	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 2427bp / cargo 8000bp (30% utilized)
- No tissue overlap (disease: ['retina'], vector: ['hematopoietic'])
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (retinal_phototransduction vs immune_hematopoietic)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010